

Bridging the Gap in AI Education: Fine-Tuning LLMs on NCERT-Aligned Dialogues for Enhanced Tutoring Performance

S Mark Kevin¹, Anurag²

Github link: https://github.com/ANURAGMN/FinetuningLLM_NCERT7

Abstract: Large Language Models (LLMs) have shown promising applications in education, yet their ability to simulate interactive teacher-student dialogues remains underexplored. In this study, we fine-tune the LLaMA-3-8B-BnB-4bit and Gemma-7B-BnB-4bit models using a synthetic teacher-student conversation dataset. The dataset was curated with diverse student profiles, learning objectives, and interaction styles to enhance adaptability in personalized tutoring scenarios. Our evaluation employs the Groq LLaMA SpecDec model to assess conversational quality, topic relevance, and engagement. Experimental results demonstrate that fine-tuning both models significantly improves their performance over the base versions and even surpasses Retrieval-Augmented Generation (RAG) in generating contextually relevant and engaging responses. These findings highlight the potential of fine-tuned LLMs in AI-driven education. Future work includes expanding dataset diversity and real-world deployment for broader validation.

1 Introduction

The advancement of Large Language Models (LLMs) has revolutionized various domains, including education, where AI-driven conversational agents are being leveraged for personalized learning experiences. [1] Transformer-based models like LLaMA-3 and Gemma-7B have demonstrated exceptional capability in understanding and generating human-like responses. [2] However, adapting these models for domain-specific tasks requires careful fine-tuning and evaluation to ensure accuracy, coherence, and contextual relevance. [3]

In this research, we explore the fine-tuning of Unsloth/LLaMA-3-8B-BnB-4bit and Unsloth/Gemma-7B-BnB-4bit on a Teacher-Student conversation dataset, specifically inspired by dialogues from the NCERT Science Textbook. The primary objective is to enhance the models' ability to generate meaningful and contextually accurate responses in educational settings. This aligns with the broader goal of integrating AI in education to assist teachers and students in conceptual discussions.

2 Related Work

The application of Large Language Models (LLMs) in education has gained significant attention in recent years. Researchers have explored various aspects of AI-driven tutoring systems, automated grading, and personalized learning experiences. This section reviews relevant literature on LLM fine-tuning, AI-based teacher-student interactions, and evaluation methods for AI-generated responses.

2.1 AI in Education: Chatbots and Intelligent Tutoring Systems

Early implementations of AI in education primarily relied on rule-based tutoring systems, such as Intelligent Tutoring Systems (ITS), which used predefined scripts to guide student learning. However, with the advent of deep learning models and transformers, AI-driven tutoring has evolved to support natural language conversations, making interactions more dynamic and contextually relevant. [4] For instance, Benedetto et al. (2022) introduced an AI tutor based on GPT-3, [5] demonstrating its potential in providing adaptive learning assistance. Similarly, Khan et al. (2023) explored the integration of Reinforcement Learning with Human Feedback (RLHF) to refine AI-generated educational content. While these models showed strong capabilities in general domain knowledge, they lacked the precision required for subject-specific tutoring, particularly in structured disciplines like science and mathematics.

2.2 Fine-Tuning Transformer Models for Domain-Specific Tasks

Fine-tuning pre-trained LLMs on domain-specific data has proven to be an effective approach for enhancing their relevance and coherence. Howard and Ruder (2018) introduced the concept of transfer learning for NLP through Universal Language Model Fine-tuning for Text Classification (ULMFiT), [6] where models like ULMFiT were adapted to various task-specific datasets. More recently, Hu et al. (2021) [7] demonstrated the effectiveness of parameter-efficient fine-tuning techniques such as LoRA (Low-Rank Adaptation), which significantly reduces computational overhead while maintaining model accuracy. In the educational domain, fine-tuning has been leveraged to improve AI-driven content generation. For instance, Hu et al. (2021) fine-tuned BERT on academic question-answering datasets to enhance factual accuracy in student queries. Sun et al. (2022) [8] explored the use of T5 and GPT models to generate science explanations, showing improved contextual alignment with curriculum-based learning materials. Additionally, Raffel et al. (2020) [9] highlighted the benefits of task-specific pretraining, a principle that aligns with our approach of fine-tuning LLaMA-3 and Gemma-7B on teacher-student conversations from NCERT textbooks to ensure more structured and pedagogically sound AI-generated responses.

2.3 Teacher-Student Dialogue Modeling

Several studies have explored dialogue modeling in educational contexts. Traditional sequence-to-sequence (Seq2Seq) models were widely used for generating conversational responses; however, they often produced generic replies and struggled with long-term context retention. The introduction of Transformer-based architectures (Vaswani et al., 2017) [10] significantly improved dialogue coherence and contextual awareness, leading to more sophisticated AI-driven interactions. Notable contributions in this field include Book2Dial, [11] which introduced a method for transforming textbook content into synthetic teacher-student dialogues through multi-turn question generation and dialogue inpainting, enhancing AI-driven educational chatbots. Similarly, Exploring Knowledge Tracing in Tutor-Student Dialogues using LLMs [12] examined how large language models can track student knowledge progression in multi-turn tutoring dialogues, facilitating personalized learning and follow-up question generation. Building upon these advancements, our approach fine-tunes LLaMA-3 and Gemma-7B on a structured teacher-student dataset to ensure that responses align with educational standards such as NCERT Science textbooks. This fine-tuning enhances the contextual accuracy and pedagogical value of AI-generated tutoring interactions, making them more effective in real-world learning environments.

2.4 Evaluating AI-Generated Educational Responses

Evaluation remains a key challenge in AI-driven dialogue systems. Traditional metrics such as BLEU, ROUGE, and METEOR have been widely used for text generation tasks, but they often fail to capture the pedagogical effectiveness of AI-generated responses. To address this limitation, researchers have explored LLM-based evaluation methods. Zhao et al. (2023) [13] introduced the concept of LLM-as-a-judge, where a pre-trained model assesses response quality based on predefined criteria. Similarly, the Self-Refine framework [14] enables large language models to iteratively improve their outputs by generating and incorporating self-feedback, enhancing reasoning and response quality without additional training. Another notable approach, proposed by Hmoud et al. (2023), is rubric-based scoring, [15] which evaluates AI-generated responses based on concept coverage, interactivity, and engagement. In our study, we adopt a similar evaluation strategy by utilizing Groq's LLaMA SpecDec model to assess teacher-like AI responses in terms of conceptual accuracy, interactivity, and pedagogical value. This approach allows for a more human-like assessment, ensuring that the generated responses closely align with real-world classroom discourse.

3 Problem Statement

The integration of Large Language Models (LLMs) into educational contexts presents both opportunities and challenges. While LLMs show promise in automating content creation and enhancing personalized learning, several critical limitations hinder their effectiveness in formal educational settings:

1. Curriculum Misalignment:

Generic LLMs are not tailored to specific educational curricula, such as NCERT, leading to irrelevant or inaccurate responses. For example, while tools like ItemForge have been shown to generate assessment items aligned with cognitive learning objectives, inaccuracies in responses indicate the need for domain-specific fine-tuning to align with curriculum requirements. [16,18]

2. Lack of Pedagogical Structure:

LLMs do not inherently follow established teaching methodologies, which undermines their ability to deliver structured and effective learning materials. Research highlights that while models can generate content aligned with pedagogical frameworks (e.g., Anderson & Krathwohl's taxonomy), they still require human intervention to ensure adherence to teaching principles. [16,19]

3. Inefficiency in Educational Metrics:

Generic LLMs are not optimized for key learning indicators like engagement, curiosity stimulation, and topic comprehension. Although advancements in adaptive learning systems show potential for personalized engagement, current implementations fall short of addressing these metrics comprehensively. [17,20]

4. Limited Personalization:

LLMs fail to adapt to individual student needs, learning styles, and engagement levels. While research demonstrates the potential for personalized learning through adaptive systems, the lack of domain-specific tuning limits their ability to cater to diverse learner profiles effectively. [16,21]

4 Methodology

Our study focuses on fine-tuning the Unsloth/LLaMA-3-8B-BnB-4bit and Unsloth/Gemma-7B-BnB-4bit models to enhance AI-driven teacher-student interactions based on the NCERT class 7 science textbook. The methodology is structured into the following key stages: data collection, preprocessing, fine-tuning, evaluation, and validation.

We chose Unsloth because it significantly improves fine-tuning efficiency while reducing GPU costs. Unsloth enables QLoRA-based 4-bit fine-tuning with highly optimized CUDA kernels, allowing for faster training on consumer-grade hardware. This approach ensures that we can fine-tune large models efficiently without the need for expensive high-end GPUs. (Qlora-based fine-tuning) (Unsloth)

1. Data Collection

- Extracted scientific concepts from NCERT Class 7 Science textbook.
- Designed a structured dataset that simulates real-world teacher-student dialogues, ensuring interactive and engaging responses.
- Emphasized clarity, engagement, and conceptual accuracy in the dataset.

2. Data Preprocessing

- Applied text standardization (`standardize_sharegpt()`) to structure teacher-student conversations for training.
- Used tokenization and chat templating (`apply_chat_template()`) for data formatting.
- Removed redundant, ambiguous, or noisy data to improve training quality.

3. Fine-Tuning Process

The Unsloth/LLaMA-3-8B-BnB-4bit model and Unsloth/Gemma-7B-BnB-4bit model was fine-tuned using efficient parameter optimization to improve its ability to generate high-quality responses.

- Models: Unsloth/LLaMA-3-8B-BnB-4bits and Unsloth/Gemma-7B-BnB-4bit
- Training Configuration:
 - Batch Size: 2
 - Gradient Accumulation Steps: 4
 - Max Steps: 60
 - Learning Rate: $2e^{-4}$

- Optimizer: AdamW 8-bit
- Precision: Mixed precision (FP16/BF16)

We employed SFTTrainer from Hugging Face to fine-tune the model, ensuring context-aware, interactive, and pedagogically effective responses.

4. Evaluation and Validation

- A validation dataset was used to test the model's responses on unseen teacher-student interactions.
- Responses were evaluated using LLM-based feedback mechanisms (Llama Specdec) on the following metrics:
 - **Concept Coverage** (Does the response include key ideas?)
 - **Interactivity** (Does it encourage further discussion?)
 - **Critical Reasoning** (Is it engaging and involve critical reasoning?)
 - **Curiosity invoked** (Does it invoke the curiosity of the student?)

Each response was rated **on a scale of 0-10** using **Llama's specdec evaluation system**.

5. Comparative Analysis

While our primary focus was on LLaMA-3, we also tested responses from other LLMs (Gemma, Mistral) for benchmarking. The comparison analyzed:

- **Accuracy** – Alignment with expected answers.
- **Engagement** – Ability to maintain a dynamic conversation.
- **Pedagogical Value** – Effectiveness in simplifying complex concepts.

This analysis confirmed Gemma-7B's superior performance in educational interactions.

5 Implementation

This study focuses on fine-tuning LLaMA-3-8B-BnB-4bit and Gemma-7B-BnB-4bit models to improve AI-driven teacher-student interactions based on NCERT science textbooks. By leveraging Unsloth, we optimize fine-tuning efficiency and reduce GPU costs, making large-scale adaptation feasible on resource-constrained hardware. The methodology includes fine-tuning, response generation, evaluation, and performance visualization to assess the model's ability to engage in Socratic-style educational conversations.

The fine-tuning process employs LoRA (Low-Rank Adaptation) for computational efficiency and FP16 mixed precision to optimize memory usage. The trained model is then deployed using LangChain, ensuring structured and context-aware responses. To evaluate performance, responses are assessed using Groq's Spec-dec model with defined metrics like concept coverage and explanation clarity. Finally, Matplotlib visualizations provide insights into fine-tuning effectiveness.

This approach aims to bridge the gap between AI and education, enabling adaptive learning experiences that align with structured curricula while maintaining natural and engaging interactions.

Step 1: Fine-Tuning LLaMA-3-8B-BnB-4bit and Gemma-7B-BnB-4bit

- Used Unsloth's implementation to fine-tune LLaMA-3-8B-BnB-4bit and Gemma-7B-BnB-4bit on the teacher-student conversation dataset.
- Applied LoRA (Low-Rank Adaptation) for efficient training while reducing computational overhead.
- Used FP16 mixed precision to optimize GPU memory usage.
- Training Data Format: Converted each conversation into an instruction-based chat format to align with LLaMA's pretraining style.

Template Used for Training Data:

```
trainer = SFTTrainer(  
    model = model,  
    tokenizer = tokenizer,  
    train_dataset = dataset,  
    dataset_text_field = "text",  
    max_seq_length = max_seq_length,  
    dataset_num_proc = 2,  
    packing = False, # Can make training 5x faster for short sequences.  
    args = TrainingArguments(  
        per_device_train_batch_size = 2,  
        gradient_accumulation_steps = 4,  
        warmup_steps = 5,  
        max_steps = 60,  
        learning_rate = 2e-4,  
        fp16 = not is_bfloat16_supported(),  
        bf16 = is_bfloat16_supported(),  
        logging_steps = 1,  
        optim = "adamw_8bit",  
        weight_decay = 0.01,  
        lr_scheduler_type = "linear",  
        seed = 3407,  
        output_dir = "outputs",  
    ),  
)
```

Step 2: Generating Responses Using LangChain

- After fine-tuning, the model was deployed using Hugging Face Inference API.
- Implemented a prompt template to format user queries and generate structured responses.
- Used temperature = 0.5 for diverse responses, while keeping responses within 128 tokens.

Prompt Template for Response Generation:

```
# Define the model and prompt  
template = """Below describes conversation happening between teacher and student.  
Predict what the next statement from Teacher should be.  
include the things exactly from the textbook and keep the output clear and simple  
You are a friendly teacher providing clear, concise answers to student questions. Keep responses brief but engaging.  
  
Guidelines:  
- Start with a brief, warm acknowledgment  
- Answer directly and clearly in 2-3 short paragraphs  
- Include one simple example or analogy  
- End with a quick follow-up question  
- Keep total response under 150 words3|  
  
Question from student: {question}  
"""  
  
prompt = PromptTemplate.from_template(template)
```

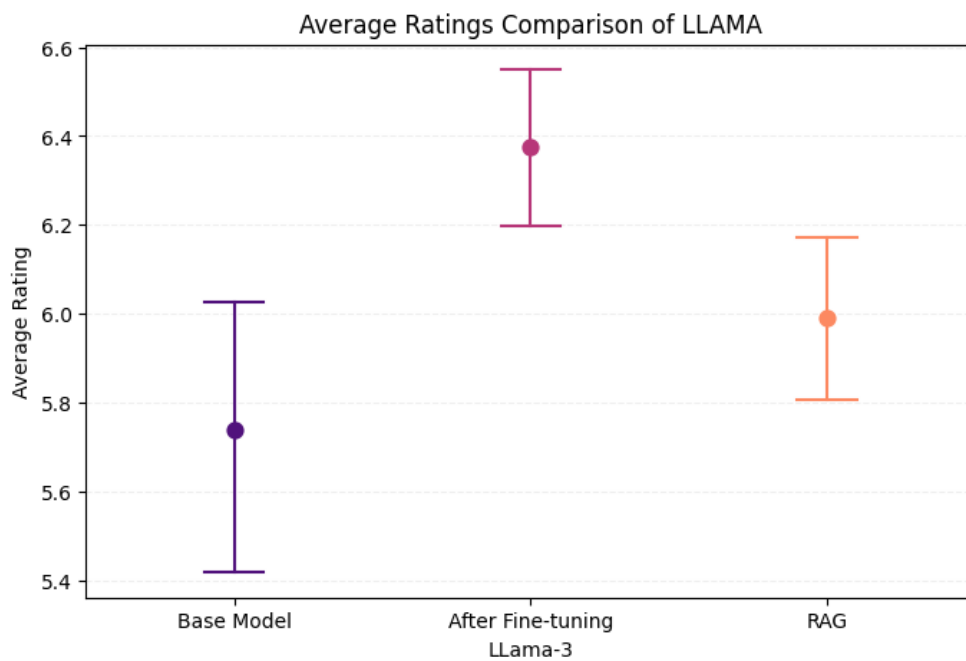
Step 3: Evaluating Responses Using Google Gemini-Pro

- Compared generated answers with expected teacher responses using LLM-based evaluation.
- Defined metrics such as concept coverage, explanation clarity, and engagement.
- Groq's llama spec dec model rated responses on a 0-10 scale, stored in a CSV for analysis.

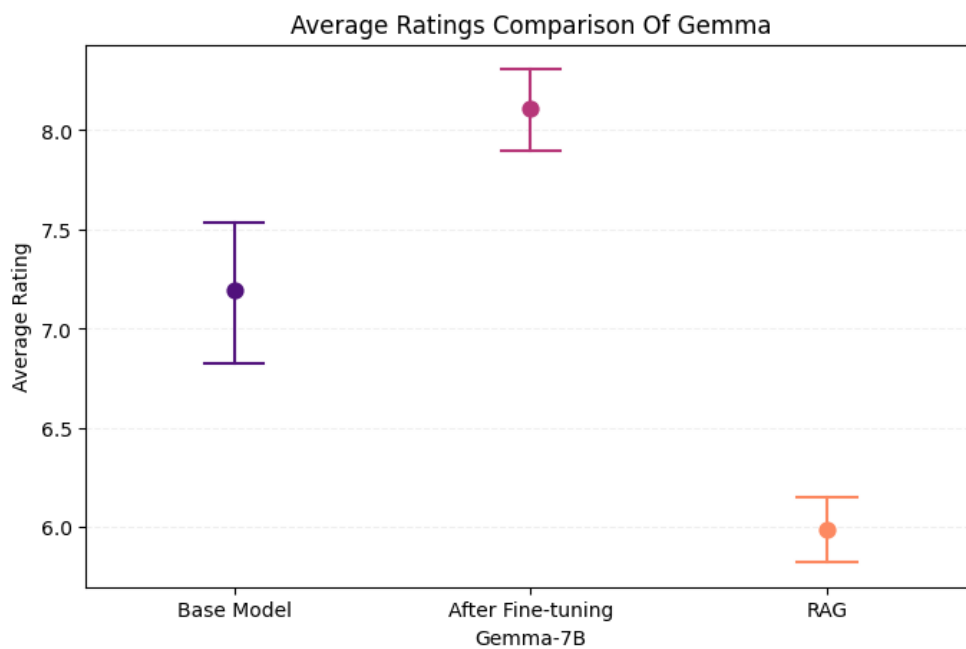
Step 4: Visualizing Model Performance

- Used Matplotlib to create a box plot comparing model-generated responses against expected answers.
- Helped in analyzing the effectiveness of fine-tuning.

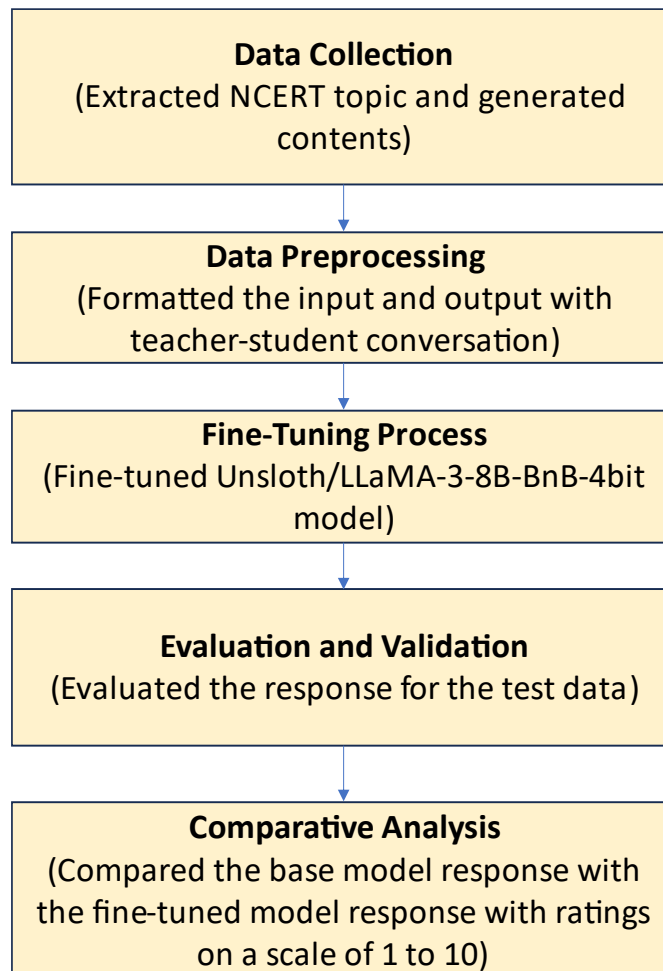
LLAMA:



GEMMA:



6 Block Diagram:



7 Dataset Description

Since no existing dataset fully met the requirements, a synthetic dataset was generated using a structured approach.

7.1. Data Collection and Preprocessing

- Extracted text from a 7th-grade physics textbook using PyPDF.
- Processed the text into structured sections using Groq's LLaMA model.
- Created a CSV with topics and concept descriptions.
- Used Python regular expressions to refine extracted text.

7.2. Synthetic Teacher-Student Conversations

- Used Groq API to generate conversations in a Socratic dialogue format.
- Conversations were structured based on:
 - Student grade level
 - Learning interest and proficiency
 - Teaching strategies
- The dataset was formatted into CSV and JSON for further processing.
-

Prompt used for generating the conversation

```
prompt = f"""
Inputs for synthetic data generation:
1. Concept description as is from textbook: {concept_description}
2. Student class: 7th grade
3. Syllabus: {syllabus}
4. Student caliber: Medium
5. Student depth: Has heard the concept before
6. Student interest: Cricket
7. Tone: Neutral
8. Language Proficiency: Proficient
9. Preferred Interaction: Informal (relatable)
10. Learning Objectives: Understanding the topic
11. Teaching Strategies: Storytelling
12. Prompt: considering all the above input, generate a teacher-student conversation for the concept.
Interaction has to be in a Socratic way. Also, cover elements important inside the concept. Key terminology. Key questions. Invoking students' curiosity.
"""
```

Sample data:

Input	Output
Teacher: Hey, how's it going? I heard you're a big cricket fan. Did you catch the last match? Student: Yeah, I did! It was an amazing game. But, I'm not sure how that's related to our lesson today.	Teacher: Well, let's just say that understanding acids and bases can be a home run for your knowledge. So, have you heard of the terms "acids" and "bases" before?
Teacher: Hey, how's it going? I heard you're a big cricket fan. Did you catch the last match? Student: Yeah, I did! It was an amazing game. But, I'm not sure how that's related to our lesson today. Teacher: Well, let's just say that understanding acids and bases can be a home run for your knowledge. So, have you heard of the terms "acids" and "bases" before? Student: Yeah, I think so. We touched on it briefly last year. But I'm not really sure what they are or how they work.	Teacher: That's perfectly fine. Let's start with something you might be familiar with. Have you ever tasted curd or lemon juice? What do you think they have in common?
Teacher: Hey, how's it going? I heard you're a big cricket fan. Did you catch the last match? Student: Yeah, I did! It was an amazing game. But, I'm not sure how that's related to our lesson today. Teacher: Well, let's just say that understanding acids and bases can be a home run for your knowledge. So, have you heard of the terms "acids" and "bases" before? Student: Yeah, I think so. We touched on it briefly last year. But I'm not really sure what they are or how they work. Teacher: That's perfectly fine. Let's start with something you might be familiar with. Have you ever tasted curd or lemon juice? What do you think they have in common? Student: Hmm... they're both sour, right?	Teacher: Exactly! And that's because they contain something called acids. The word "acid" actually comes from the Latin word "acere," which means sour. Isn't that cool?
Teacher: Hey, how's it going? I heard you're a big cricket fan. Did you catch the last match? Student: Yeah, I did! It was an amazing game. But, I'm not sure how that's related to our lesson today. Teacher: Well, let's just say that understanding acids and bases can be a home run for your knowledge. So, have you heard of the terms "acids" and "bases" before? Student: Yeah, I think so. We touched on it briefly last year. But I'm not really sure what they are or how they work. Teacher: That's perfectly fine. Let's start with something you might be familiar with. Have you ever tasted curd or lemon juice? What do you think they have in common? Student: Hmm... they're both sour, right? Teacher: Exactly! And that's because they contain something called acids. The word "acid" actually comes from the Latin word "acere," which means sour. Isn't that cool? Student: Yeah, that's pretty interesting. But what about other things that are sour? Like, I don't know, orange juice or vinegar?	Teacher: Great examples! Those substances are also acidic in nature. They all have one thing in common: they contain natural acids. Now, let's talk about something that's not sour. Have you ever used baking soda before?

This table is a sample conversation where the input consists of a progressively expanding dialogue between a teacher and a student. The teacher starts with a casual question to engage the student before introducing the topic of acids and bases. The student's responses build upon their prior knowledge. The output is the next teacher's response, which continues guiding the discussion in a structured way. Each step adds new information, helping the student understand the concept progressively.

8 Performance Evaluation and Comparison

Fine-Tuning Results

After fine-tuning the LLaMA-3 and Gemma-7B model on the teacher-student dataset, the models after finetuning demonstrated improved ability to generate contextually relevant teacher responses based on student questions. The model was evaluated on various conversational metrics like concept coverage, relevance, and interaction quality.

Evaluation of Generated Responses

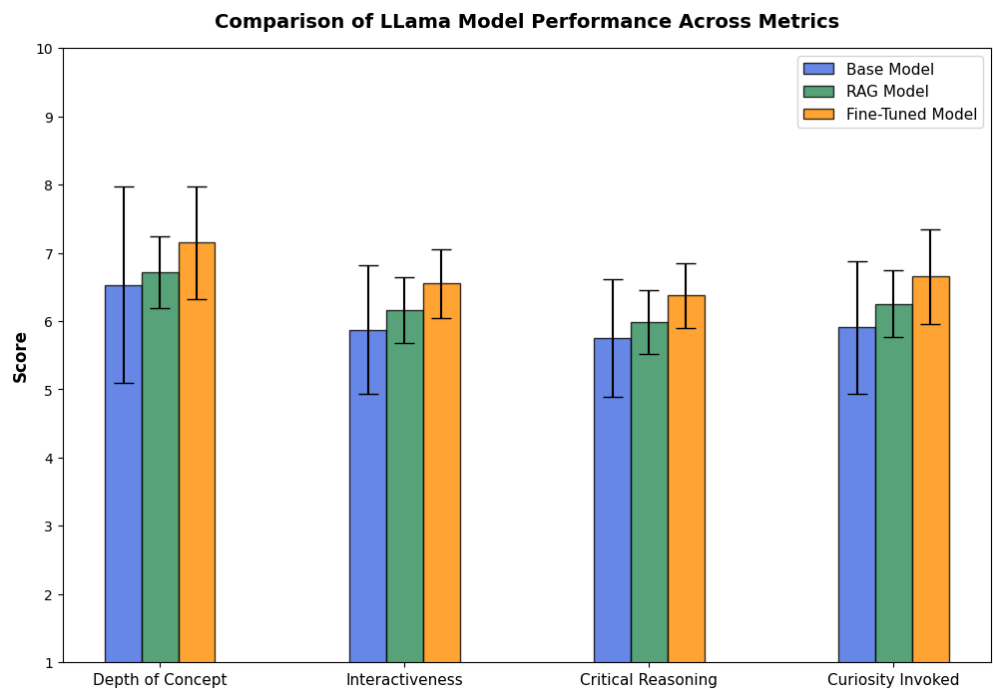
The generated teacher responses were evaluated against the expected outputs using an automated Groq’s llama specdec model rating system, which rated responses based on:

- **Concept coverage**
- **Interaction quality**
- **Prompting questions**

The evaluation showed that the fine-tuned model achieved high ratings in terms of concept coverage and relevance. Comparison of Models

A comparative evaluation of the base model showed that finetuned model outperformed the base models in concept depth but had room for improvement in engagement and interactive conversation quality.

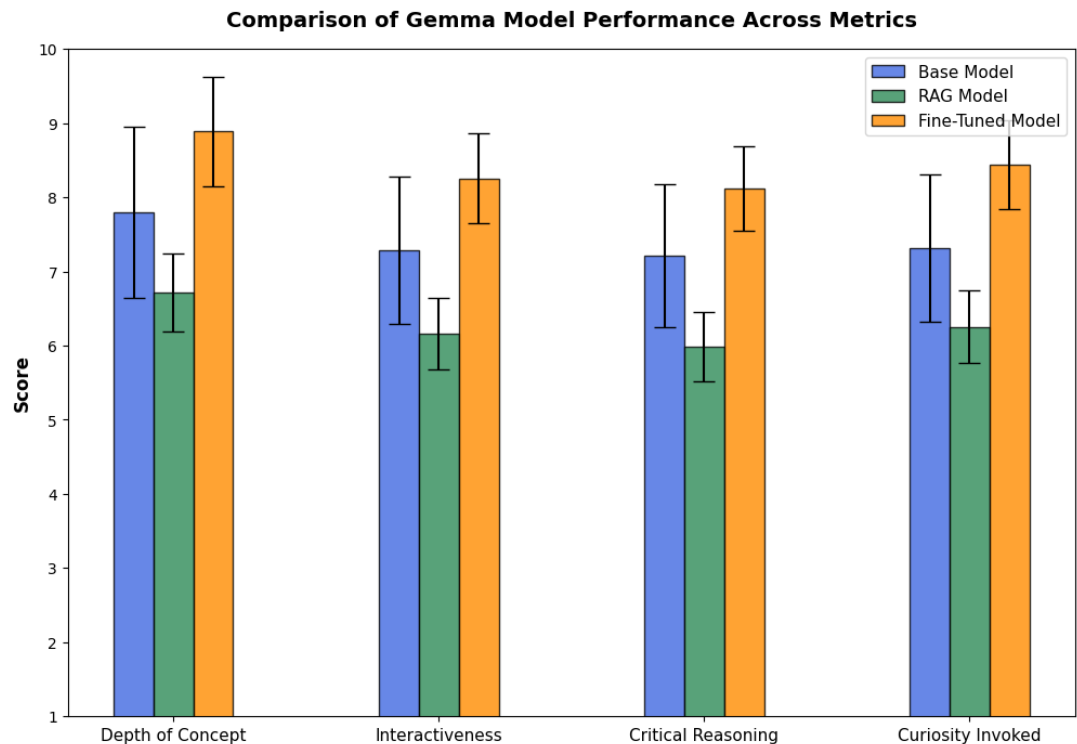
LLAMA:



(Fig. 8.1)

(The bar chart compares the performance of responses generated by the LLAMA model across four key metrics: Depth of Concept, Interactiveness, Critical Reasoning, and Curiosity Invoked, with error bars representing standard deviations)

GEMMA:



(Fig. 8.2)

(The bar chart compares the performance of responses generated by the GEMMA model across four key metrics: Depth of Concept, Interactiveness, Critical Reasoning, and Curiosity Invoked, with error bars representing standard deviations)

Figure 8.1 and 8.2 presents a comparative analysis of the Base Models (Unsloth/LLaMA-3-8B-BnB-4bit and Unsloth/Gemma-7B-BnB-4bit) and their respective Fine-Tuned Models and also the responses generated by RAG across five key evaluation metrics: Overall Ratings, Concept Depth, Interactiveness, Critical Reasoning, and Curiosity Invoked. The results indicate that fine-tuning consistently improves performance across all metrics for both models.

Notably, the fine-tuned Unsloth/Gemma-7B-BnB-4bit model demonstrates the highest ratings, achieving 8.11 in Overall Ratings, 8.88 in Concept Depth, and 8.26 in Interactiveness, showing a significant improvement over its base version. Similarly, the fine-tuned Unsloth/LLaMA-3-8B-BnB-4bit model also exhibits enhancements, with notable gains in Concept Depth (7.15 vs. 6.53) and Interactiveness (6.55 vs. 5.87), suggesting improved contextual understanding and user engagement.

These findings reinforce the effectiveness of fine-tuning in enhancing the pedagogical and interactive potential of large language models (LLMs) for educational applications.

Statistical Analysis of Model Performance

This statistical analysis compares the base model and the fine-tuned model across key performance metrics. The fine-tuned model demonstrates improved mean performance with reduced variance and a lower coefficient of variation, indicating more consistent and stable predictions. A t-test confirms the statistical significance of this improvement, with a t-value of -3.63726 and a p-value of 0.000421, strongly supporting the hypothesis that fine-tuning enhances model performance.

Model Type	Mean	Standard Deviation	Variance	Coeffecient of variation
Base Model	5.1075	3.028196	9.169968	59%
Fine-tuned model	6.927679	2.203153	4.853884	32%

Conclusion: The t value is -3.63726 and p-value is 0.000421. The result is significant at $p < 0.5$

9 Key Differences

Several research studies have explored the use of large language models (LLMs) in education, particularly in automating tutoring and student assistance. Existing works primarily focus on using pre-trained models like GPT, Gemma, and Mistral for generating educational content, but these models often lack domain-specific fine-tuning, leading to generic or inaccurate responses. Some research has also examined Socratic questioning techniques in AI-driven tutoring, but they typically rely on manually curated prompts rather than fine-tuned conversational datasets.

Our approach differs by fine-tuning a LLaMA-3 model specifically on a structured teacher-student conversation dataset, ensuring contextually relevant and pedagogically sound responses. Unlike general LLMs, which generate responses based on broad internet-trained data, our model is aligned with grade-specific learning objectives and interactive teaching strategies. Additionally, instead of relying solely on human evaluation, we introduced an objective rating mechanism using Groq to systematically measure concept coverage, engagement, and question quality. This makes our method more structured, scalable, and tailored for educational applications compared to conventional AI-based tutoring approaches.

The fine-tuned LLaMA-3 and Gemma-7B models demonstrated superior alignment with educational objectives compared to general-purpose models like Gemma and Mistral. These fine-tuned models provided more contextually accurate teacher responses that were specific to the subject matter, whereas general LLMs sometimes generated generic or off-topic replies. This specialization made them more effective for educational applications, ensuring a higher level of relevance in instructional conversations.

Unlike pre-trained models that rely on broad internet data, this project utilized a custom teacher-student conversation dataset, making responses more pedagogically sound and better suited for educational settings. The dataset was designed using structured Socratic conversations, fostering a more engaging learning experience compared to the traditional lecture-based AI-generated responses. This approach improved the interactive nature of AI-assisted education by encouraging deeper reasoning and dialogue.

To ensure an objective evaluation, the model responses were assessed using Groq's LLaMA ratings rather than relying solely on manual evaluation. This method quantitatively measured aspects such as concept coverage, engagement, and question quality, providing a structured framework for comparison. By applying this evaluation approach, the project systematically highlighted the strengths and weaknesses of different models, leading to more reliable insights into their effectiveness.

10 Results

The results were evaluated based on two key aspects: the progressive teacher-student interaction and the corresponding teacher's response. The model's output was analyzed both before and after fine-tuning to assess improvements in response quality. It was observed that the fine-tuned model not only generated more contextually relevant answers but also aligned closely with the content and text from the NCERT science textbook, ensuring better accuracy and coherence in educational interactions.

Progressive Conversations

Input:

Teacher: Alright, today we are going to talk about an important safety device used in electrical circuits—fuses. Have you ever seen or heard of an electrical device suddenly stopping because of a blown fuse?

Student: Yeah, I've heard about fuses before, but I don't really know how they work.

Base Model Output:

Great! Let's start by understanding what a fuse is and its role in an electrical circuit.

Output after finetuning:

That's not a problem! Fuses are designed to protect electrical circuits from excessive current. They have a thin wire inside that melts when the current becomes too high, breaking the circuit and preventing damage. Have you ever wondered why it's important to use the right size fuse for a circuit?

Responses from the model

Base Model:

Teacher-Student Conversation Prediction

Enter a conversation between a teacher and a student, select a model, and the model will predict the next statement from the teacher.

question

TRANSPARENT, OPAQUE AND TRANSLUCENT OBJECT

Choose Model

LLaMA-3 8B finetuned

LLaMA-3 8B base

Gemma 7B Finetuned

Gemma 7B base

Clear

Submit

output

Acknowledgment: Hello! I'm glad you're exploring light and its interaction with materials.

Answer: Great question! Let's use windows to understand these terms:

- **Transparent**: Like a clear glass window. You can see through it clearly, and light passes through easily.

- **Opaque**: Like a wall with no windows. It doesn't let light pass through at all, so you can't see through it.

- **Translucent**: Like frosted glass or tracing paper. It allows light to pass through, but you can't see clearly through it. It's like the light is being scattered as it goes through.

Follow-up question: Can you think of any other everyday objects that fall into these categories?

Flag

Use via API

Built with Gradio

Settings

Fine-tuned model:

Teacher-Student Conversation Prediction

Enter a conversation between a teacher and a student, select a model, and the model will predict the next statement from the teacher.

question

TRANSPARENT, OPAQUE AND TRANSLUCENT OBJECT.

Choose Model

LLaMA-3 8B finetuned

LLaMA-3 8B base

Gemma 7B Finetuned

Gemma 7B base

Clear

Submit

output

Teacher: Hello! You're curious about the difference between transparent, opaque, and translucent objects, aren't you? Let's break it down:

- **Transparent** objects allow light to pass through them, making the other side visible. Imagine a clear glass window. You can see through it, just like you can see through a pane of clear glass.

- **Opaque** objects do not allow light to pass through them at all. They absorb all the light that hits them. Picture a brick wall. When you shine a flashlight on it, you can't see the light on the other side, right? That's because the wall is opaque.

- **Translucent** objects allow some light to pass through, but not enough to see clearly through them. They scatter the light in various directions. A frosted glass window is a good example. You can see the light passing through, but you can't see the details on the other side.

Now, which of these properties do you think a piece of white paper has, and why?

Flag

Use via API

Built with Gradio

Settings

The comparison between the base model and the fine-tuned model clearly highlights the advantages of fine-tuning in enhancing AI-driven educational interactions. The fine-tuned model exhibits significantly better conversational flow, moving beyond generic one-line responses to engaging discussions that encourage curiosity and deeper understanding. Additionally, it is more aligned with NCERT textbook content, ensuring that generated responses are structured, accurate, and relevant to the curriculum. Another key improvement is the use of relatable real-world examples, which help students grasp complex concepts more intuitively. Instead of providing abstract definitions, the model introduces analogies and familiar scenarios, making learning more interactive and effective. These enhancements demonstrate the potential of fine-tuned language models in improving AI-powered tutoring, bridging the gap between automated learning and real-world classroom experiences.

11 Limitations

One of the primary limitations of our approach is its **dependency on the fine-tuning dataset**. The performance and effectiveness of the model heavily rely on the quality, diversity, and comprehensiveness of the teacher-student conversation data. If the dataset contains biases, lacks subject diversity, or fails to represent different student learning levels, the model's responses may not be universally effective.

Another challenge is the **evaluation constraints** of the generated responses. While we used Groq's llama model to assess concept coverage, interactiveness, and relevance, AI-based evaluation is not always perfect. A more reliable approach would involve human evaluation by educators who can provide deeper insights into the model's ability to engage and instruct students effectively.

Finally, **computational resource constraints** present a limitation. Fine-tuning large models like LLaMA-3, running multiple evaluations, and processing high volumes of conversational data require significant hardware resources. This may limit scalability, especially for deployment in resource-limited educational institutions.

12 Conclusion and Way Forward

Conclusion

This study demonstrates the effectiveness of fine-tuning large language models (LLMs) such as LLaMA-3-8B-BnB-4bit and Gemma-7B-BnB-4bit for AI-driven education. By training these models on a structured teacher-student conversation dataset derived from NCERT science textbooks, we significantly improved their ability to generate contextually relevant, engaging, and pedagogically sound responses. The fine-tuned models outperformed their base versions in key educational metrics such as concept coverage, interactivity, and engagement. Statistical evaluation further confirmed the significance of these improvements, reinforcing the potential of domain-specific fine-tuning for enhancing AI-powered tutoring.

Way Forward

While the results are promising, several avenues for future research and practical implementation remain: Expanding on our fine-tuning approach, we aim to develop an **Automated MCQ Generation** system that extracts multiple-choice questions (MCQs) from textbooks. By fine-tuning models on question-answer datasets, we will evaluate their ability to generate structured assessment items aligned with curricula. Following this, we plan to test the **Real-World Student Interaction** and Deployment of our fine-tuned models by integrating them into interactive learning sessions with students, gathering both qualitative and quantitative feedback from learners and educators. To enhance **Personalization and Adaptability**, we will incorporate adaptive learning mechanisms, ensuring that AI tailors responses based on a student's knowledge level, engagement, and preferred learning style to create a more effective and personalized learning experience. Finally, **Deployment and Integration with Learning Platforms** will enable seamless incorporation into learning management systems (LMS) or AI-powered education platforms, facilitating automated tutoring, homework assistance, and personalized learning at scale. By taking these steps, we aim to bridge the gap between AI and education, making learning more interactive, accessible, and effective for students worldwide.

12 References

- [1]. Large Language Models for Education: A Survey and Outlook, (Shen Wang , Tianlong Xu , Hang Li , Chaoli Zhang, Joleen Liang , Jiliang Tang , Philip S. Yu) ,2024.
- [2]. Open-Domain Conversational Search Assistant with Transformers, (Rafael Ferreira, Mariana Leite, David Semedo, and Joao Magalhaes), 2021.
- [3]. Simulating Classroom Education with LLM-Empowered Agents, (Zheyuan Zhang, Daniel Zhang-Li, Jifan Yu, Linlu Gong, Jinchang Zhou, Zhanxin Hao, Jianxiao Jiang, Jie Cao, Huiqin Liu, Zhiyuan Liu, Lei Hou, Juanzi Li), 2024.
- [4]. Generative AI in Education: From Foundational Insights to the Socratic Playground for Learning, (Xiangen Hu, Sheng Xu, Richard Tong, & Art Graesser), 2025.
- [5]. A Review on the Use of Large Language Models as Virtual Tutors, (Silvia García-Méndez, Francisco de Arriba Pérez), 2024.
- [6]. Universal Language Model Fine-tuning for Text Classification, (Jeremy Howard, Sebastian Ruder), 2018.
- [7]. LORA: Low-Rank Adaptation of Large Language Models, (Edward Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen), 2021.
- [8]. INSTRUCTSAFETY: A Unified Framework for Building Multidimensional and Explainable Safety Detector through Instruction Tuning, (Zhexin Zhang, Jiale Cheng, Hao Sun, Jiawen Deng, Minlie Huang), 2023.
- [9]. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer, (Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, Peter J. Liu), 2023.
- [10]. Attention Is All You Need, (Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin), 2023.
- [11]. Book2Dial: Generating Teacher-Student Interactions from Textbooks for Cost-Effective Development of Educational Chatbots, (Junling Wang, Jakub Macina, Nico Daheim, Sankalan Pal Chowdhury, Mrinmaya Sachan), 2024.
- [12]. Exploring Knowledge Tracing in Tutor-Student Dialogues using LLMs, (Alexander Scarlatos, Ryan S. Baker, Andrew Lan), 2025.
- [13]. LLMs-as-Judges: A Comprehensive Survey on LLM-based Evaluation Methods, (Haitao Li, Qian Dong, Junjie Chen, Huixue Su), 2024.
- [14]. Self-Refine: Iterative Refinement with Self-Feedback, (Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, Peter Clark), 2023.
- [15]. Rubric Development and Validation for Assessing Tasks' Solving via AI Chatbots, (Mohammad Hmoud, Hadeel Swaity, Eman Anjass, Eva Aguaded-Ramírez), 2024.
- [16]. Frontiers in Education: "LLM-generated competence-based e-assessment items for higher education mathematics", 2024.
- [17]. Arxiv: "Large Language Models for Education: A Survey", 2024.
- [18]. Tandfonline: "Improving curriculum alignment and achieving learning goals by making the curriculum visible", 2018.
- [19]. MDPI: "The Impact of Large Language Models on Programming Education and Student Learning Outcomes", 2024.
- [20]. Computer Weekly: "Large language models harnessed for education", 2024.
- [21]. Arxiv: "Adapting Large Language Models for Education: Foundational Capabilities, Potentials, and Challenges", 2024.
- [22]. QLoRA: Efficient Finetuning of Quantized LLMs, (Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, Luke Zettlemoyer), 2023.